

CHE 1105

ORGANIC CHEMISTRY LABORATORY

B.Sc. General Degree

First Year Semester II

Department of Physical Sciences

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PRACTICAL ORGANIC CHEMISTRY

The procedure used to identify and characterize organic compounds is known as Qualitative Organic Analysis. It is an important part of practical organic chemistry. During this course, you will learn the appropriate methods required to establish the class to which an organic compound belongs and to identify the functional groups present in that compound. Spectroscopy is probably the most powerful and modern tool available to the chemist for determining the structure of an unknown compound. It is often possible to determine the structure using only spectroscopic techniques. But qualitative organic analysis gives you the chance to learn and understand the chemistry of different functional groups.

CLASSIFICATION OF ORGANIC COMPOUNDS

1. TESTS FOR ELEMENTS

Carbon, hydrogen, oxygen, nitrogen, sulfur and the halogens are elements most commonly found in organic compounds. Phosphorus may be present to a lesser extent. Detection of the elements present in the unknown sample is necessary because this provides information on the functional group present in the compound and hence is a guide to the types of tests to be carried out for each sample. For example, the absence of nitrogen means the absence of amines, amides, ammonium salts and nitro compounds and thus tests for those functional groups need not be performed.

Carbon and hydrogen are assumed to be present in all organic compounds. There is no specific test for oxygen. Tests given below are used to detect the presence of nitrogen, sulfur, halogens and phosphorus in an unknown compound. It is also necessary to find out whether those elements present in the compound are in the ionic or non-ionic forms.

1.1 Preliminary tests

Beilstein Test

Compounds containing halogens give a green flame when heated on a small piece of copper gauze. This is due to volatile copper halides formed on the copper. Some organic compounds containing nitrogen but not containing halogens (eg- urea, thiourea, pyridine derivatives) also give a positive test due to the formation of volatile copper cyanide under these conditions. Therefore a green flame indicates that halogen is probably present and a negative result (no green flame) means the definite absence of halogens.

Procedure: Heat a small piece of copper gauze in a Bunsen flame until it oxidized and there is no green flame. Dip the gauze in the sample and heat it again in the flame and observe the color. A positive test is indicated by a green flame.

Ignition Test

Procedure: Place a small amount of the compound on a spatula and place it in the flame of the Bunsen burner.

Compounds giving a sooty flame have a high degree of unsaturation and may be aromatic. These compounds contain little oxygen, and have a high carbon-to-hydrogen ration. They burn

at a low temperature with a yellow flame (e.g. naphthalene). Much carbon is produced when they are burned. Compounds that contain oxygen generally burns at a high temperature with a clean blue flame (e.g. Ethanol).

Sodium Fusion (Lassaigne fusion) Test

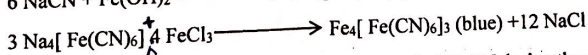
Nitrogen, sulfur, halogens and phosphorous present in organic compounds are converted to NaCN, Na₂S, NaX (X= Cl, Br or I) and Na₃P respectively by fusion of the organic compounds with molten sodium.

Procedure: Place a little of the organic compound at the bottom of ignition tube and a small piece of sodium on the upper wall of the ignition tube. Heat the piece of sodium, first gently and then strongly in a Bunsen flame, until it melts and allow the molten sodium to drop on to the organic compound. Heat the tube to red hot for 1-2 minutes to insure the complete reaction and plunge the red hot tube in to distilled water (20 ml) contained in a boiling tube. Stir and warm the contains of the tube and filter to obtained a clear alkaline solution of the sodium salts. {Do not empty broken glass into the sink}

a. **Test for sulfur;** add sodium nitroprusside, Na₂ [Fe (CN)₅NO] solution to the sodium fusion extract . An intense purple color due to [Fe (CN)₅NOS] complex is obtained when sulfur is present .

b. **Test for nitrogen ;** Cyanide present in the extract is detected by converting it to the Ferro cyanide , which gives a blue precipitate with Fe⁺³

Procedure: Add a freshly prepared FeSO₄ to the sodium fusion extract. When sulfur is absent a green precipitate of Fe (OH)₂ is formed .If sulfur is present a black precipitate of FeS is also formed. Boil gently to ensure the formation of Ferro cyanide and to oxidize some of the Fe⁺² to Fe⁺³. Cooled and acidify with dilute H₂SO₄. A blue precipitate or a blue coloration indicates the presence of nitrogen. If a blue precipitate / color is not seen, the solution should be filtered before recording the absence of nitrogen. If very little of the precipitate is formed, this will be already seen as blue spots on the filter paper.



If sulfur is present, NaCNS may be formed instead of NaCN during the sodium fusion. This will give FeCNS with Fe⁺³ giving a red color. In such cases, sodium fusion should be repeated taking more sodium to ensure the conversion of initially formed NaCNS to NaCN.



c. **Tests for halogens**

Cyanide and sulfide ions formed during the sodium fusion interfere with the test for halides because AgCN and Ag₂S may be precipitated. Therefore these ions have to be removed first.

Procedure: Acidify the sodium fusion extracts with dilute HNO_3 and boils it for about 2 minutes. This will drive off any HCN or hydrogen sulphide that is formed. When the solution cools, add Silver nitrate AgNO_3 solution. A white precipitate readily soluble in NH_4OH indicates the presence of Cl^- , Br^- is indicated by an off-white precipitate, which is sparingly soluble in NH_4OH and I^- by a yellow precipitate insoluble in NH_4OH .

Differentiation of Cl^- , Br^- and I^-

Acidify the extract with dil. HNO_3 and boil for 2 mins. Cool the solution, add Carbon tetrachloride (About 0.5ml) or Dichloromethane and a few drops of chlorine water. Shake the contents of the tube vigorously and set aside to allow the layers to separate. An orange to brown color in the organic layer indicates Br^- and violet indicates I^- . Absence of a color indicates Cl^- .

d. Test for Phosphorus; Phosphide (P^{3-}) present in the sodium fusion extract is detected by converting it Phosphate (PO_4^{3-}).

Procedure: Boil 1ml of sodium fusion extract with excess conc. HNO_3 to oxidize Phosphide (P^{3-}) to Phosphate (PO_4^{3-}). Cool, add an equal volume of Ammonium Molybdate solution and warm the mixture. A yellow precipitate of Ammonium Phosphomolybdate indicates the presence of Phosphorus in the test sample.

✓ Ionic or non-ionic nature of Sulphur, Halogens and Phosphorus

An aqueous solution of the original sample is used for this test. Place the sample in a test tube, add distilled water, shake and carry out the following tests with the supernatant.

- i. **Halogens** - Add excess dil. HNO_3 and add AgNO_3 solution. A precipitate indicates the presence of halide ions.
- ii. **Sulphur** - Acidify with dil. HNO_3 and add BaCl_2 solution. A white precipitate indicates the presence of sulphur as SO_4^{2-} .
- iii. **Phosphorous** - Acidify with conc. HNO_3 , add ammonium molybdate solution and warm. A yellow precipitate indicates the presence of phosphorous as PO_4^{3-} .

2. SOLUBILITY BEHAVIOUR.

The solubility of an organic compound provides useful information regarding its structure and properties. In general, acidic compounds are soluble in bases and basic compounds are soluble in acids. Neutral compounds which are insoluble in water are also insoluble in both acids and bases.

Following charts indicate solvents in which compounds containing various functional groups are likely to dissolve.